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Numerical Methods

AAS203D

One-day revision notes: definitions, solved examples, code and important questions

Program: B.Tech | **Branches:** CSE, IT, CSE-AIML, CSE-DS, AIDS | **Semester:** 2 | **Credits:** 4

L-T-P: 3-1-0 | **Scheme:** AKTU B.Tech 1st year, CSE stream, 2026-27 onwards

UNIT 1 Ordinary Differential Equations

UNIT 2 Complex Analysis

UNIT 3 Laplace Transform

UNIT 4 Fourier Series and Fourier Transform

UNIT 5 Applied Linear Algebra

Original revision notes written for the AKTU syllabus. In spite of its name, this second-semester maths paper for the CSE stream covers differential equations, complex analysis, Laplace and Fourier methods and applied linear algebra. Every unit has the same parts:

1. **2-mark definitions and formulas**: short answers and formulas you can write directly in Section A.
2. **Core concepts**: methods and step-by-step procedures.
3. **Solved examples**: worked problems of the type examiners set, with every step.
4. **Key points to remember**.
5. **Important questions**: the long questions to prepare, with what your answer must contain.

How to use these notes the day before the paper

- **Morning (3 hours):** Unit 1 (linear ODEs with constant coefficients, particular integrals, Cauchy-Euler, variation of parameters) and Unit 3 (Laplace transform, inverse, convolution, solving ODEs). These two units together give many sure marks.
- **Afternoon (3 hours):** Unit 2 (analytic functions, Cauchy-Riemann, Milne-Thomson, Cauchy's integral formula, residues, real integrals).
- **Evening (2 hours):** Unit 4 (Fourier series, half-range series, Parseval, Fourier transform) and Unit 5 (EVD, SVD, LU, QR, linear maps, kernel and range, norms), then the "Last-hour checklist".
- **In the exam:** write the rule for each step (for example "PI = e^(ax)/f(a) when f(a) ≠ 0"), draw the contour and mark the poles for residue problems, and sketch the function before a Fourier series.

AKTU paper pattern (theory, 70 marks, 3 hours): Section A has 7 short questions of 2 marks, all compulsory (14). Section B has 5 questions of 7 marks, attempt any 3 (21). Section C has 5 questions (one per unit), each with two parts, attempt one part of each, 7 marks each (35). Every unit gets its own Section C question, so do not skip any unit. Internal assessment carries another 30 marks.

UNIT 1 Ordinary Differential Equations

✍️ 2-mark definitions and formulas

- **Linear ODE of order n with constant coefficients:** $(D^n + a_1D^{n-1} + \ldots + a_n)\, y = X(x)$, where $D = d/dx$; written $f(D)\, y = X$.
- **Complete solution:** $y = CF + PI$, where CF (complementary function) solves $f(D)y = 0$ and PI (particular integral) is any one solution of $f(D)y = X$.
- **Auxiliary equation (AE):** $f(m) = 0$, obtained by replacing D with m.
- **CF rules:**
- Distinct real roots m_1, m_2: $c_1e^{(m_1x)} + c_2e^{(m_2x)}$.
- Repeated root m: $(c_1 + c_2x)e^{(mx)}$.
- Complex roots $\alpha \pm i\beta$: $e^{(\alpha x)}(c_1 \cos \beta x + c_2 \sin \beta x)$.
- **PI for $X = e^{(ax)}$:** $e^{(ax)}/f(a)$ if $f(a) \neq 0$; if $f(a) = 0$, $x\, e^{(ax)}/f'(a)$.
- **PI for X = sin ax or cos ax:** replace D^2 by $-a^2$ in f(D) (if non-zero).
- **PI for $X = x^m$:** expand $[f(D)]^{-1}$ in ascending powers of D.
- **PI for $X = e^{(ax)}\, V$:** $e^{(ax)} \cdot [1/f(D + a)]\, V$ (shifting rule).
- **Cauchy-Euler equation:** $x^n\, y^n + a_1\, x^{n-1}\, y^{n-1} + \ldots + a_n\, y = X$; reduced to constant coefficients by $x = e^t$ ($x\, D = \theta$, $x^2D^2 = \theta(\theta - 1)$).
- **Wronskian:** $W = y_1y_2' - y_2y_1'$; non-zero for independent solutions.

💡 Core concepts

Variation of parameters (second order $y'' + P\, y' + Q\, y = R$):

1. Find the CF: $y = c_1y_1 + c_2y_2$.
2. Compute $W = y_1y_2' - y_2y_1'$.
3. $PI = -y_1 \int (y_2R/W)\, dx + y_2 \int (y_1R/W)\, dx$.

It works for any R (sec x, tan x, cosec x), where the short-cut rules fail.

Second-order equations with variable coefficients:

- **Change of independent variable:** for $y'' + P\, y' + Q\, y = R$, put $z = z(x)$ chosen so that $(dz/dx)^2 = Q/a^2$ (a constant). The equation becomes $d^2y/dz^2 + P_1\, dy/dz + Q_1\, y = R_1$, with:
- $P_1 = (z'' + P\, z')/z'^2$.
- $Q_1 = Q/z'^2$.
- $R_1 = R/z'^2$.
- It is useful if P_1 comes out constant.
- **Normal form (removal of first derivative):** put $y = u\, v$ with $v = e^{(-\frac{1}{2}\int P\, dx)}$. The equation becomes $u'' + I\, u = S$, with:
- $I = Q - P^2/4 - P'/2$.
- $S = R/v$.
- Useful if I is constant or simple.

Cauchy-Euler steps:

1. Put x = e^t, so x d/dx = θ = d/dt.
2. Rewrite and solve the constant-coefficient equation in t.
3. Replace t by log x.

🧮 Solved examples

Example 1 (constant coefficients). Solve $y'' - 3y' + 2y = e^{(3x)}$.

- AE: $m^2 - 3m + 2 = 0$, m = 1, 2. CF = $c_1e^x + c_2e^{(2x)}$.
- PI = $e^{(3x)}/f(3) = e^{(3x)}/(9 - 9 + 2) = e^{(3x)}/2$.
- **$y = c_1e^x + c_2e^{(2x)} + e^{(3x)}/2$.**

Example 2 (variation of parameters). Solve $y'' + y = \sec x$.

- CF = $c_1 \cos x + c_2 \sin x$; $y_1 = \cos x$, $y_2 = \sin x$; $W = \cos^2x + \sin^2x = 1$.
- PI = $-\cos x \int \sin x \sec x \, dx + \sin x \int \cos x \sec x \, dx = -\cos x (-\log \cos x) + \sin x (x)$.
- **$y = c_1 \cos x + c_2 \sin x + x \sin x + \cos x \cdot \log \cos x$.**

Example 3 (Cauchy-Euler). Solve $x^2y'' - xy' + y = \log x$.

- x = e^t: $[\theta(\theta - 1) - \theta + 1] y = t$, that is $(\theta - 1)^2 y = t$.
- CF = $(c_1 + c_2t)e^t$. PI = $(1 - \theta)^{-2} t = (1 + 2\theta + \ldots) t = t + 2$.
- **$y = (c_1 + c_2 \log x) x + \log x + 2$.**

Example 4 (PI with sin). Find the PI of $(D^2 + 4)y = \sin 3x$.

- Replace D^2 by −9: PI = $\sin 3x/(-9 + 4) =$ **$-(1/5) \sin 3x$**.

Key points to remember

- CF from AE roots (distinct, repeated, complex).
- PI rules: e^(ax), sin/cos ($D^2 \to -a^2$), x^m (expand), e^(ax)V (shift), failure case (multiply by x, differentiate f).
- Variation of parameters: W and the two integrals.
- Cauchy-Euler: x = e^t, θ(θ − 1).
- Change of independent variable; normal form $I = Q - P^2/4 - P'/2$.

⭐ Important questions

1. Solve linear ODEs with constant coefficients for exponential, trigonometric and polynomial right-hand sides.
2. Solve a Cauchy-Euler (homogeneous) linear equation.
3. Solve by the method of variation of parameters (sec x, tan x, cosec x type).
4. Solve a second-order equation by changing the independent variable.
5. Reduce an equation to normal form and solve it.

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UNIT 2 Complex Analysis

✍️ 2-mark definitions and formulas

- **Complex function:** $w = f(z) = u(x, y) + i\,v(x, y)$, with $z = x + iy$.
- **Limit and continuity:** as for real functions, but the limit must be the same along every path in the plane.
- **Differentiable:** $f'(z) = \lim (f(z + \Delta z) - f(z))/\Delta z$ exists independent of the direction of Δz.
- **Analytic (holomorphic) function:** differentiable at every point of a region.
- **Cauchy-Riemann (CR) equations, Cartesian:** $u_x = v_y$ and $u_y = -v_x$.
- **CR equations, polar:** $u_r = (1/r)\, v_\theta$ and $v_r = -(1/r)\, u_\theta$.
- **Harmonic function:** satisfies Laplace's equation $u_{xx} + u_{yy} = 0$; the real and imaginary parts of an analytic function are harmonic (and called harmonic conjugates).
- **Singularity:** a point where f is not analytic. It may be isolated, removable, a pole of order m, or essential.
- **Residue:** the coefficient of $1/(z - a)$ in the Laurent expansion about a. For a simple pole, $\text{Res} = \lim (z - a)\, f(z)$; for a pole of order m, $\text{Res} = [1/(m - 1)!] \lim d^{m-1}/dz^{m-1} [(z - a)^m f(z)]$.
- **Cauchy's integral theorem:** if f is analytic inside and on a closed curve C, then $\oint_C f(z)\, dz = 0$.

💡 Core concepts

Necessary and sufficient conditions for analyticity: the CR equations hold and the partial derivatives are continuous. Then $f'(z) = u_x + i\, v_x$.

Milne-Thomson method (find f(z) from u or v):

1. Given u: find $u_x = \varphi_1(x, y)$, $u_y = \varphi_2(x, y)$.
2. $f'(z) = \varphi_1(z, 0) - i\, \varphi_2(z, 0)$ (put $x = z$, $y = 0$).
3. Integrate: $f(z) = \int [\varphi_1(z, 0) - i\varphi_2(z, 0)]\, dz + C$.

If v is given instead, use $f'(z) = \psi_1(z, 0) + i\, \psi_2(z, 0)$, with $\psi_1 = v_y$ and $\psi_2 = v_x$.

Complex integration: $\int_C f(z)\, dz = \int (u\, dx - v\, dy) + i \int (v\, dx + u\, dy)$.

Cauchy's integral formula: for f analytic inside and on C and a inside C:

- $f(a) = (1/2\pi i) \oint f(z)/(z - a)\, dz$.
- $f^{(n)}(a) = (n!/2\pi i) \oint f(z)/(z - a)^{n+1}\, dz$.

Taylor and Laurent series:

- **Taylor (f analytic in $|z - a| < R$):** $f(z) = \Sigma\, f^{(n)}(a)(z - a)^n/n!$.
- **Laurent (in an annulus $r < |z - a| < R$):** $f(z) = \Sigma\, a_n(z - a)^n + \Sigma\, b_n(z - a)^{-n}$ (the principal part).
- **Method:** use partial fractions and geometric series, choosing the expansion that converges in the given region.

Residue theorem: $\oint_C f(z)\, dz = 2\pi i \times$ (sum of residues at poles inside C).

Evaluating real integrals:

- **Type 1, $\int_0^{2\pi} F(\cos\theta, \sin\theta)\, d\theta$:**
- Put $z = e^{i\theta}$: $\cos\theta = (z + 1/z)/2$, $\sin\theta = (z - 1/z)/(2i)$, $d\theta = dz/(iz)$.
- Integrate over $|z| = 1$.
- **Type 2, $\int_{-\infty}^{\infty} P(x)/Q(x)\, dx$** (degree of Q at least 2 more than P, no real poles): take a large semicircle in the upper half-plane; the integral equals $2\pi i \times$ (sum of residues in the upper half-plane).

🧮 Solved examples

Example 5 (analytic check). Show that $f(z) = z^2$ is analytic and $f(z) = \bar{z}$ is not.

- $z^2 = (x^2 - y^2) + i(2xy)$: $u_x = 2x = v_y$, $u_y = -2y = -v_x$. CR holds everywhere and the partials are continuous, so z^2 is **analytic**, with $f' = 2x + i2y = 2z$.
- $\bar{z} = x - iy$: $u_x = 1$, $v_y = -1$, so $u_x \neq v_y$; $\bar{z}$ is **nowhere analytic**.

Example 6 (Milne-Thomson). Find the analytic function with $u = e^x \cos y$.

- $u_x = e^x \cos y$, $u_y = -e^x \sin y$. At $x = z$, $y = 0$: $\varphi_1 = e^z$, $\varphi_2 = 0$.
- $f'(z) = e^z$, so **$f(z) = e^z + C$** ($v = e^x \sin y$).

Example 7 (residue theorem). Evaluate $\oint_C dz/(z^2 + 1)$, where C is $|z - i| = 1$.

- Poles at $z = \pm i$; only $z = i$ lies inside C.
- Res at $i = 1/(2i)$. Integral $= 2\pi i \times 1/(2i) = \boldsymbol{\pi}$.

Example 8 (real integral, type 1). Evaluate $\int_0^{2\pi} d\theta/(2 + \cos\theta)$.

- With $z = e^{i\theta}$: integral $= \oint dz/(iz(2 + (z + 1/z)/2)) = (2/i) \oint dz/(z^2 + 4z + 1)$.
- Poles at $z = -2 \pm \sqrt{3}$; only $-2 + \sqrt{3}$ is inside $|z| = 1$. Residue $= 1/(2\sqrt{3})$.
- Integral $= (2/i)(2\pi i)(1/(2\sqrt{3})) = \boldsymbol{2\pi/\sqrt{3}}$.

Example 9 (real integral, type 2). $\int_{-\infty}^{\infty} dx/(x^2 + 1)$: pole at i in the upper half-plane, residue $1/(2i)$, integral $= 2\pi i/(2i) = \boldsymbol{\pi}$.

Key points to remember

- CR: $u_x = v_y$, $u_y = -v_x$; polar $u_r = v_\theta/r$, $v_r = -u_\theta/r$.
- Analytic → u, v harmonic; Milne-Thomson: $f'(z) = u_x(z, 0) - i\, u_y(z, 0)$.
- Cauchy theorem ($\oint = 0$) and integral formula ($f(a)$, $f^{(n)}(a)$).
- Taylor vs Laurent; types of singularities.
- Residue formulas; $\oint = 2\pi i \sum \text{Res}$; real integrals by $z = e^{i\theta}$ and semicircle.

⭐ Important questions

1. Derive the Cauchy-Riemann equations (Cartesian and polar) and test analyticity.
2. Show that a given function is harmonic and find its harmonic conjugate and f(z) by Milne-Thomson.
3. State and use Cauchy's integral formula to evaluate integrals.
4. Expand a function in Taylor and Laurent series in given regions; classify singularities.

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5. State the residue theorem and evaluate real definite integrals using contour integration.

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UNIT 3 Laplace Transform

✍️ 2-mark definitions and formulas

- **Laplace transform:** $L\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\, dt$.
- **Existence conditions:** f is piecewise continuous on every finite interval and of exponential order ($|f(t)| \le M e^{at}$ for large t).
- **Linearity:** $L\{a f + b g\} = a F + b G$.
- **Standard transforms:**
- $L\{1\} = 1/s$.
- $L\{t^n\} = n!/s^{n+1}$.
- $L\{e^{at}\} = 1/(s - a)$.
- $L\{\sin at\} = a/(s^2 + a^2)$.
- $L\{\cos at\} = s/(s^2 + a^2)$.
- $L\{\sinh at\} = a/(s^2 - a^2)$.
- $L\{\cosh at\} = s/(s^2 - a^2)$.
- **First shifting theorem:** $L\{e^{at} f(t)\} = F(s - a)$.
- **Second shifting theorem:** $L\{f(t - a)\, u(t - a)\} = e^{-as} F(s)$.
- **Change of scale:** $L\{f(at)\} = (1/a) F(s/a)$.
- **Multiplication by t:** $L\{t f(t)\} = -F'(s)$; generally $L\{t^n f(t)\} = (-1)^n F^{(n)}(s)$.
- **Division by t:** $L\{f(t)/t\} = \int_s^\infty F(s)\, ds$.

💡 Core concepts

Transforms of derivatives and integrals:

- $L\{f'(t)\} = sF(s) - f(0)$.
- $L\{f''(t)\} = s^2F(s) - s f(0) - f'(0)$.
- $L\{\int_0^t f(u)\, du\} = F(s)/s$.

Periodic functions: for f with period T, $L\{f\} = [1/(1 - e^{-sT})] \int_0^T e^{-st} f(t)\, dt$.

Unit step function: $u(t - a) = 0$ for $t < a$ and 1 for $t \ge a$; $L\{u(t - a)\} = e^{-as}/s$. Piecewise functions are written using unit steps.

Inverse Laplace transform: methods:

- Standard table.
- Shifting theorems.
- **Partial fractions** (most common).
- Inverse of derivatives (multiplication by t).
- **Convolution theorem:** $L^{-1}\{F(s)G(s)\} = (f * g)(t) = \int_0^t f(u)\, g(t - u)\, du$.

Solving ODEs with Laplace transform:

1. Take L of both sides using the derivative formulas and the initial conditions.

2. Solve algebraically for Y(s).
3. Use partial fractions and inverse transform to get y(t).

The initial conditions are built in, so no constants need to be found.

🧮 Solved examples

Example 10 (shifting). $L\{e^{-2t}\cos 3t\} = (s+2)/((s+2)^2+9)$.

Example 11 (multiplication by t). $L\{t\sin at\} = -\frac{d}{ds}[a/(s^2+a^2)] = \mathbf{2as/(s^2+a^2)^2}$.

Example 12 (convolution). Find $L^{-1}\{1/(s^2(s+1))\}$.

- With $F = 1/s^2$ ($f = t$) and $G = 1/(s+1)$ ($g = e^{-t}$): $\int_0^t u\,e^{-(t-u)}\,du = \mathbf{t - 1 + e^{-t}}$. (Partial fractions give the same.)

Example 13 (solving an ODE). Solve $y'' + 3y' + 2y = e^{-3t}$, $y(0) = 0$, $y'(0) = 0$.

- $(s^2+3s+2)\,Y = 1/(s+3)$, so $Y = 1/((s+1)(s+2)(s+3))$.
- Partial fractions: $Y = (1/2)/(s+1) - 1/(s+2) + (1/2)/(s+3)$.
- $\mathbf{y(t) = \tfrac{1}{2}e^{-t} - e^{-2t} + \tfrac{1}{2}e^{-3t}}$. Check $y(0) = \tfrac{1}{2} - 1 + \tfrac{1}{2} = 0$.

Example 14 (periodic function). A square wave $f(t) = 1$ for $0 < t < a$ and -1 for $a < t < 2a$ (period $2a$).

- $L\{f\} = [1/(1-e^{-2as})]\,[\int_0^a e^{-st}\,dt - \int_a^{2a} e^{-st}\,dt] = (1-e^{-as})^2/(s(1-e^{-2as}))$ = **$(1/s)\tanh(as/2)$**.

Key points to remember

- Table of standard transforms; existence conditions.
- First shift $F(s-a)$; second shift $e^{-as}F(s)$; scale $(1/a)F(s/a)$.
- $t\,f(t) \to -F'(s)$; $f(t)/t \to \int_s^\infty F$.
- $L\{f'\} = sF - f(0)$; $L\{f''\} = s^2F - s\,f(0) - f'(0)$; integral $\to F/s$.
- Periodic formula; unit step e^{-as}/s.
- Inverse: partial fractions, convolution; ODE solving steps.

⭐ Important questions

1. Define the Laplace transform and state existence conditions. Find transforms using shifting, scale and multiplication-by-t theorems.
2. Find the Laplace transform of a periodic function and of a function expressed with unit steps.
3. Find inverse Laplace transforms by partial fractions.
4. State the convolution theorem and use it to find an inverse transform.
5. Solve a linear ODE (or a system) with initial conditions by the Laplace transform.

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UNIT 4 Fourier Series and Fourier Transform

✍️ 2-mark definitions and formulas

- **Periodic function:** $f(x + T) = f(x)$ for all x; the smallest such T is the period.
- **Even and odd functions:** even if $f(-x) = f(x)$ ($\cos x$, x^2); odd if $f(-x) = -f(x)$ ($\sin x$, x).
- **Fourier series on $(-\pi, \pi)$:** $f(x) = a_0/2 + \Sigma\,(a_n \cos nx + b_n \sin nx)$, with:
- $a_0 = (1/\pi)\int f\,dx$.
- $a_n = (1/\pi)\int f \cos nx\,dx$.
- $b_n = (1/\pi)\int f \sin nx\,dx$.
- **Even function on $(-\pi, \pi)$:** $b_n = 0$ (cosine series only). **Odd function:** $a_0 = a_n = 0$ (sine series only).
- **Dirichlet's conditions:** f is single-valued, periodic, has finitely many maxima, minima and discontinuities in a period, and is absolutely integrable. Then the series converges to $f(x)$, and at a jump to the average $[f(x^+) + f(x^-)]/2$.
- **General interval $(-l, l)$:** replace nx by $n\pi x/l$; $a_n = (1/l)\int_{-l}^{l} f \cos(n\pi x/l)\,dx$, and so on.
- **Half-range series on $(0, l)$:** sine series $b_n = (2/l)\int_0^l f \sin(n\pi x/l)\,dx$ (odd extension); cosine series $a_n = (2/l)\int_0^l f \cos(n\pi x/l)\,dx$ (even extension).
- **Parseval's identity $(-\pi, \pi)$:** $(1/\pi)\int f^2\,dx = a_0^2/2 + \Sigma\,(a_n^2 + b_n^2)$.
- **Fourier transform:** $F(\omega) = \int_{(-\infty)}^{(\infty)} f(x)\, e^{(-i\omega x)}\,dx$; inverse $f(x) = (1/2\pi)\int F(\omega)\, e^{(i\omega x)}\,d\omega$.
- **Finite Fourier sine and cosine transforms on $(0, l)$:** $F_s(n) = \int_0^l f \sin(n\pi x/l)\,dx$; $F_c(n) = \int_0^l f \cos(n\pi x/l)\,dx$.

💡 Core concepts

Useful integrals: for integer n:

- $\sin n\pi = 0$.
- $\cos n\pi = (-1)^n$.
- $\int x \sin nx\,dx$ and $\int x^2 \cos nx\,dx$ are done by parts (Bernoulli's rule: $\int uv = u v_1 - u' v_2 + u'' v_3 - \ldots$).

Change of interval: a function on $(0, 2l)$ or (a, b) is handled with the substitution that maps it to the standard interval; the formulas use limits a to $a + 2l$.

Fourier integral theorem: $f(x) = (1/\pi)\int_0^{\infty} \int_{(-\infty)}^{(\infty)} f(t) \cos \omega(t - x)\,dt\,d\omega$, under Dirichlet conditions and absolute integrability. This leads to the Fourier sine and cosine integrals and transforms.

Properties of the Fourier transform:

- Linearity.
- Shifting: $f(x - a) \to e^{(-i\omega a)} F(\omega)$.
- Scaling: $f(ax) \to (1/|a|)\, F(\omega/a)$.
- Modulation.
- Convolution: transform of $f * g = F \cdot G$.
- Parseval: $\int |f|^2\,dx = (1/2\pi)\int |F|^2\,d\omega$.

Application: solving PDEs (heat, wave), signal analysis (frequency spectrum).

🧮 Solved examples

Example 15 (Fourier series of x). Find the Fourier series of f(x) = x on (−π, π).

- f is odd, so $a_o = a_n = 0$.
- $b_n = (2/\pi)\int_o^{\pi} x \sin nx\, dx = 2(-1)^{(n+1)}/n$.
- **$x = 2[\sin x - (\sin 2x)/2 + (\sin 3x)/3 - \ldots]$**.
- At x = π/2 this gives the Leibniz series π/4 = 1 − 1/3 + 1/5 − ...

Example 16 (Fourier series of x^2 and a series sum). $f(x) = x^2$ on (−π, π) is even.

- $a_o = (2/\pi)\int_o^{\pi} x^2\, dx = 2\pi^2/3$; $a_n = 4(-1)^n/n^2$.
- **$x^2 = \pi^2/3 + 4\,\Sigma\,(-1)^n \cos nx / n^2$**.
- At x = 0: $0 = \pi^2/3 + 4\Sigma(-1)^n/n^2$, so $1 - 1/4 + 1/9 - \ldots = \mathbf{\pi^2/12}$.
- At x = π: $\pi^2 = \pi^2/3 + 4\Sigma\, 1/n^2$, so $\Sigma\, 1/n^2 = \mathbf{\pi^2/6}$.

Example 17 (Parseval). Using Example 15: $(1/\pi)\int x^2\, dx = 2\pi^2/3 = \Sigma\, b_n^2 = \Sigma\, 4/n^2$, so again $\Sigma\, 1/n^2 = \mathbf{\pi^2/6}$.

Example 18 (Fourier transform). Find the Fourier transform of $f(x) = e^{(-a|x|)}$, a > 0.

- $F(\omega) = \int_{(-\infty)}^{0} e^{(ax)} e^{(-i\omega x)}\, dx + \int_o^{\infty} e^{(-ax)} e^{(-i\omega x)}\, dx = 1/(a - i\omega) + 1/(a + i\omega) = \mathbf{2a/(a^2 + \omega^2)}$.

Key points to remember

- Euler formulas a_o, a_n, b_n; even → cosine only; odd → sine only.
- Dirichlet conditions; average value at jumps.
- General interval (nπx/l); half-range sine and cosine series.
- Parseval; standard series sums ($\pi^2/6$, $\pi^2/12$, π/4).
- Fourier transform pair; $e^{(-a|x|)} \to 2a/(a^2 + \omega^2)$; finite sine and cosine transforms.

⭐ Important questions

1. Find the Fourier series of a function on (−π, π) and deduce the sum of a numerical series.
2. Find the Fourier series of a function on a general interval (−l, l) or (0, 2l).
3. Obtain half-range sine and cosine series of a function on (0, l).
4. State Parseval's identity and use it to find a series sum.
5. State the Fourier integral theorem; find the Fourier transform (or finite sine/cosine transform) of a function.

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UNIT 5 Applied Linear Algebra

✍️ 2-mark definitions and formulas

- **Eigenvalue decomposition (EVD):** $A = P\,D\,P^{-1}$, where D holds eigenvalues and P the eigenvectors; for a symmetric A, $A = Q\,D\,Q^{T}$ with orthogonal Q.
- **Singular value decomposition (SVD):** any $m \times n$ matrix $A = U\,\Sigma\,V^{T}$, where U ($m \times m$) and V ($n \times n$) are orthogonal and Σ holds the singular values $\sigma_i = \sqrt{}$(eigenvalues of $A^{T}A$) in decreasing order.
- **LU decomposition (Doolittle):** $A = LU$ with L lower triangular with 1s on the diagonal and U upper triangular.
- **QR decomposition:** $A = QR$ with Q having orthonormal columns and R upper triangular; obtained by the Gram-Schmidt process.
- **Linear transformation:** a map $T: V \to W$ with $T(au + bv) = aT(u) + bT(v)$.
- **Matrix of a linear transformation:** the matrix whose columns are the images of the basis vectors, so that $T(x) = Ax$.
- **Kernel (null space):** $\ker T = \{v : T(v) = 0\}$.
- **Range (image):** $\{T(v) : v \in V\}$.
- **Rank-nullity theorem:** $\dim(\ker T) + \dim(\text{range } T) = \dim V$.
- **Inner product space:** a vector space with an inner product $\langle u, v\rangle$ that is linear, symmetric (conjugate-symmetric for complex) and positive definite; gives length $\|v\| = \sqrt{\langle v, v\rangle}$ and angle $\cos\theta = \langle u, v\rangle/(\|u\|\|v\|)$.

💡 Core concepts

Vector norms:

- $\|x\|_1 = \Sigma|x_i|$ (Manhattan).
- $\|x\|_2 = \sqrt{(\Sigma x_i^2)}$ (Euclidean).
- $\|x\|\infty = \max|x_i|$.
- **Properties:** $\|x\| \geq 0$ (zero only for 0), $\|\alpha x\| = |\alpha|\|x\|$, and the triangle inequality $\|x + y\| \leq \|x\| + \|y\|$.
- **Cauchy-Schwarz:** $|\langle u, v\rangle| \leq \|u\|\|v\|$.

Doolittle LU method:

1. Write $A = LU$ with unknown entries; solve row by row for U and column by column for L.
2. To solve $AX = B$: first $LY = B$ by forward substitution, then $UX = Y$ by back substitution. - **Benefit:** factorize once, then solve quickly for many right-hand sides.

Gram-Schmidt process (QR):

1. $q_1 = a_1/\|a_1\|$.
2. $u_2 = a_2 - (q_1 \cdot a_2)\,q_1$; $q_2 = u_2/\|u_2\|$; and so on.
3. R has entries $r_{ij} = q_i \cdot a_j$. - **Uses:** least squares, the QR algorithm for eigenvalues.

SVD steps:

1. Find the eigenvalues of $A^{T}A$; $\sigma_i = \sqrt{\lambda_i}$.

2. The eigenvectors of A^TA give V.
3. $u_i = A\,v_i / \sigma_i$ give U. - **Uses:** data compression (keep the largest σ), principal component analysis, recommendation systems, image compression, pseudo-inverse, and the rank of A (number of non-zero σ).

Applications of EVD: powers of matrices, PCA (eigenvectors of the covariance matrix), Google PageRank, vibration modes.

🧮 Solved examples

Example 19 (LU decomposition). Factorize A = [2 3; 4 7] by Doolittle's method.

- U row 1 = [2 3]; $l_{21} = 4/2 = 2$; $u_{22} = 7 - 2 \times 3 = 1$.
- **L = [1 0; 2 1], U = [2 3; 0 1]**. Check: LU = [2 3; 4 7].

Example 20 (QR decomposition). Find the QR decomposition of A = [1 1; 1 0].

- $a_1 = (1, 1)$: $q_1 = (1, 1)/\sqrt{2}$; $r_{11} = \sqrt{2}$.
- $r_{12} = q_1 \cdot a_2 = 1/\sqrt{2}$; $u_2 = (1, 0) - (1/2)(1, 1) = (1/2, -1/2)$; $r_{22} = \|u_2\| = 1/\sqrt{2}$; $q_2 = (1, -1)/\sqrt{2}$.
- **Q = (1/√2)[1 1; 1 −1], R = [√2 1/√2; 0 1/√2]**.

Example 21 (singular values). Find the singular values of A = [3 0; 4 5].

- A^TA = [25 20; 20 25], with eigenvalues 45 and 5.
- **$\sigma_1 = \sqrt{45} = 3\sqrt{5} \approx 6.71$, $\sigma_2 = \sqrt{5} \approx 2.24$**. Both non-zero, so rank 2; $|\det A| = 15 = \sigma_1\sigma_2$.

Example 22 (kernel and range). T: $\mathbb{R}^3 \to \mathbb{R}^2$, T(x, y, z) = (x + y, y + z).

- Matrix A = [1 1 0; 0 1 1], rank 2, so the range is all of $\mathbb{R}^2$ (dim 2).
- Kernel: x + y = 0, y + z = 0 gives (t, −t, t), so ker T = span{(1, −1, 1)}, **nullity 1**.
- Rank-nullity: 2 + 1 = 3 = dim $\mathbb{R}^3$.

Example 23 (norms). For v = (3, −4, 0): $\|v\|_1 = \mathbf{7}$, $\|v\|_2 = \mathbf{5}$, $\|v\|_\infty = \mathbf{4}$.

Key points to remember

- EVD: $A = PDP^{-1}$ (QDQ^T if symmetric). SVD: $A = U\Sigma V^T$, $\sigma = \sqrt{\text{eig}(A^TA)}$.
- Doolittle LU (L with unit diagonal); forward then back substitution.
- QR by Gram-Schmidt; $r_{ij} = q_i \cdot a_j$.
- Linear map matrix; kernel, range; rank + nullity = n.
- Inner product, norms (1, 2, ∞), Cauchy-Schwarz.

⭐ Important questions

1. Find the eigenvalue decomposition of a matrix and use it to compute a power of the matrix.
2. Find the singular value decomposition of a given matrix and explain its applications.
3. Factorize a matrix by Doolittle's LU method and use it to solve a system.
4. Find the QR decomposition using the Gram-Schmidt process.

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5. Find the matrix, kernel and range of a linear transformation and verify rank-nullity. Define an inner product space and vector norms.

Last-hour checklist

☐ CF and PI rules; Example 1 (e^(3x)/2); variation of parameters (x sin x + cos x log cos x); Cauchy-Euler (log x + 2); change of independent variable; normal form.

☐ CR (Cartesian and polar); harmonic; Milne-Thomson (e^z); Cauchy integral formula; Taylor and Laurent; residues; $\oint = \pi$; $\int d\theta/(2 + \cos\theta) = 2\pi/\sqrt{3}$; $\int dx/(x^2 + 1) = \pi$.

☐ Laplace table and theorems; $2as/(s^2 + a^2)^2$; convolution (t − 1 + e^(−t)); ODE solution (½e^(−t) − e^(−2t) + ½e^(−3t)); periodic square wave (1/s) tanh(as/2); unit step.

☐ Fourier coefficients; x series; x^2 series; $\pi^2/6$ and $\pi^2/12$; half-range; Parseval; Fourier transform $2a/(a^2 + \omega^2)$.

☐ EVD; SVD (3√5, √5); LU ([1 0; 2 1], [2 3; 0 1]); QR (Gram-Schmidt); kernel (1, −1, 1); rank-nullity; norms (7, 5, 4).

Books prescribed by AKTU

These are the books named in the official AKTU syllabus. Buy or borrow them through a bookshop or your college library. The WorldCat links show which libraries hold a copy.

#	Title	Author(s)	Publisher	Where to get it legally
1	Advanced Engineering Mathematics	Erwin Kreyszig	John Wiley & Sons	WorldCat
2	Higher Engineering Mathematics	B. V. Ramana	Tata McGraw-Hill	WorldCat
3	Higher Engineering Mathematics	B. S. Grewal	Khanna Publishers	WorldCat
4	Advanced Engineering Mathematics	R. K. Jain, S. R. K. Iyengar	Narosa Publishing House	WorldCat
5	Complex Variables and Applications (reference)	James Ward Brown, Ruel V. Churchill	McGraw-Hill	WorldCat
6	Fourier Series and Boundary Value Problems (reference)	James Ward Brown, Ruel V. Churchill	McGraw-Hill	WorldCat
7	Laplace Transforms, Schaum's Outline (reference)	Murray R. Spiegel	McGraw-Hill	WorldCat
8	Linear Algebra: A Modern Introduction (reference)	David Poole	Brooks/Cole	WorldCat

Syllabus source: AKTU B.Tech 1st year CSE stream course structure and syllabus, 2026-27 (official PDF)